

Use of ICTs during ongoing protracted socio-political disruptions.

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ABSTRACT

Human-Computer Interaction (HCI) research has increasingly focused on enhancing human development in diverse contexts through technology. However, regions going through ongoing socio-political unrest face varied developmental challenges. My dissertation explores the impact of socio-political conflict in Kashmir, India, on critical domains of human development, including education, employment, and mental health. The ongoing conflict has disrupted education, leading to prolonged closures of schools and colleges and exacerbated unemployment, particularly among young people with higher education levels with significant mental health challenges. Through qualitative methods, my work draws on field visits and semi-structured interviews to investigate the barriers in education, job-seeking, and access to professional mental health support. Based on the findings, my work provides socio-technical design recommendations shedding light on the impact of protracted socio-political crises on human development and contributes to the HCI communities understanding of such regions.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Crisis Informatics, Disruption, Technology, Culture, Internet shutdown, Kashmir, India.

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1 INTRODUCTION

The research in Human-Computer Interaction (HCI) has been extensively directed toward understanding the impact of technology on various developmental domains of human life. For example, education [3, 50], employment [55], entrepreneurship [22], mental health [38], and women's health and well-being [26], among others in resourced and under-resourced settings. Similarly, international organizations have recognized these domains and identified them

as crucial for achieving the United Nations Sustainable Development Goals (UN SDGs) [49]. However, regions affected by ongoing socio-political conflicts present a unique set of challenges for HCI research. These regions often have restricted access to technology and resources, and the socio-political context can significantly impact the way technology is used and the outcomes it produces. Thus, HCI research in such regions requires a specialized approach considering the region's specific context and challenges.

My dissertation, situates the work in Kashmir, India, affected by socio-political conflict for more than three decades. Historically, the region of Jammu & Kashmir (J&K)¹, particularly Kashmir in India, has been a disputed territory between India and Pakistan since 1947. The dispute has resulted in several wars and numerous military conflicts, leading to a complex political situation in the region [23]. The dispute has resulted in several wars and numerous military conflicts, leading to a complex political situation that impacts critical areas of human development in the region. This has significantly affected the education sector, resulting in prolonged closures of schools and colleges [54]. Consequently, the employment sector has also been adversely affected, given the challenging educational infrastructure and the existing ecosystem in the state. Moreover, the abrogation of Article 370² in August 2019 led to a division of J&K into two union territories, triggering a year-long lockdown and the world's longest internet shutdown. These disruptions have contributed to a rising unemployment rate, especially among young people aged 18-29, those with higher levels of education, and individuals with technical or postgraduate degrees. In addition to these challenges, the prolonged conflict in the region has led to various mental health issues among its residents, including anxiety, depression, and post-traumatic stress disorder (PTSD) [20, 21]. A significant portion of the population in the Kashmir Valley, approximately 45%, suffers from these mental health issues, with particularly high prevalence rates compared to the global average [20]. Access to mental health resources remains limited, with only 10% of those affected receiving treatment due to a lack of available services [20]. Addressing these issues in the context of the ongoing conflict presents an opportunity for designing technology solutions that are sensitive to the unique needs of the population.

With this, my research explores the barriers to various critical domains essential for human development, e.g., education and job-seeking opportunities in the region of J&K, India, which has been affected by socio-political conflict. Through field visits and semi-structured interviews with diverse stakeholders and non-government organizations. This study examines the impact of disruption on education and the role of community learning centers in overcoming education barriers within the context of the conflict. The study also examines the systemic infrastructural challenges

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¹[https://en.wikipedia.org/wiki/Jammu_and_Kashmir_\(union_territory\)](https://en.wikipedia.org/wiki/Jammu_and_Kashmir_(union_territory))

²https://en.wikipedia.org/wiki/Article_370_of_the_Constitution_of_India

faced by youth in the region in their job-seeking efforts and analyzes the workarounds used to mitigate these challenges. I also explore how do the people in the region navigate the access to support seeking for mental health and wellbeing given the intertwined contextual challenges of the region. The research contributes to filling the gap in understanding the impact of socio-political conflict on education, employment, and mental health well-being in this specific context and provides socio-technical design recommendations for similar contexts. Further, I plan to investigate the effect on the population's entrepreneurship experiences and contribute to the growing conversations in the field of HCI with regions facing a protracted socio-political crisis.

2 RELATED WORK

Below, I outline the prior scholarship in HCI, which is at the intersection of crisis informatics and the relationship of technology around education, employment, and mental health well-being for development work.

2.1 Education, Employment and HCI

Research in Human-computer interaction (HCI) and Computer-supported collaborative work (CSCW) has explored the use of technology for education and employment in both resourced and under-resourced contexts. A large body of existing scholarship in HCI, CSCW, and ICTD has focused on technology-mediated education, for example designing language learning applications [24], MOOCs [34], social-robots [3, 27], virtual reality-based learning experience [52], etc to assist students as well as teachers [50]. The studies have also covered a wide array of design techniques to explore education and technology, for example, participatory design [14, 58], cooperative inquiry [15], asset-based design [6, 31], etc. Moreover, the prior research has explored technology design to support student's education within the classroom [1, 47] and out-of-school learning environments [29, 39] leveraging the existing technological assets within the communities to improve the learning outputs.

Moreover, the field of HCI and CSCW research has also sought to understand the challenges faced by job seekers from diverse backgrounds to develop effective and inclusive technologies. Studies have identified a variety of challenges faced by these job seekers, including economic disadvantage [11, 56], low digital literacy and accessibility to essential resources [40, 45, 55, 57], and societal norms that restrict job search activities [7, 13]. These challenges may manifest in the form of poor social capital [11, 19], lack of self-efficacy and interpersonal skills [18, 56], limited mentorship [16, 48], biases in online recruitment [28], and limited experience with technology [35]. Technology-mediated solutions such as Massive Open Online Courses (MOOCs), online Question & Answer platforms (QnA), indigenous communication platforms, and community groups on social media [4, 11] have been previously studied and suggested to support job seekers. Online platforms such as LinkedIn and Glassdoor have also emerged as popular means of job search and recruitment [12]. However, research on these technologies has mostly been conducted in Western settings and may not fully apply to non-Western contexts. The digital divide, which refers to the difficulties users face accessing online information

due to internet connectivity and infrastructure gaps, is often overlooked in existing research. This divide is especially prominent in non-normative situations, like conflict-affected areas such as Kashmir, where access to information and intermittent internet are significant challenges. Therefore, studying socio-cultural and socio-political factors in different regions is crucial to designing technologies that assist processes relating to education and employment.

2.2 Mental Health and HCI

The field of Human-Computer Interaction (HCI) is increasingly focusing on mental health well-being and its related provisions, as evidenced by various studies [37, 38]. With this, digital mental health tools are emerging as potential solutions to address mental health challenges, with numerous studies exploring their potential and application in providing effective care and support. For example, they have been utilized for early detection of mental health disorders [46], health monitoring [32], online therapy [53], support groups [25], and psycho-education [2]. Additionally, HCI research in mental health has employed ecological perspectives investigating the roles of caregivers and family members [9], peer networks [5], socio-cultural [37], and socio-political [44] influence in an individual's mental health care journey. Despite these advancements in mental health care support, individuals still struggle with mental health issues, particularly in areas with limited resources and support [36]. Furthermore, individuals in regions with prolonged socio-political conflict situations face obstacles to seeking mental health care and assistance. The existing efforts to address mental health care in such regions become inadequate to meet the needs of the individuals in such regions. As each conflict situation is distinct from another, especially when considering the diverse geographical contexts. Our research underscores these challenges in the context of J&K, India, a region of territorial dispute for decades that has experienced prolonged armed conflict and socio-political unrest [33]. According to research reports, there are increased rates of depression, anxiety, and post-traumatic stress disorder (PTSD) [20, 21] among the population. Mental health is a critical concern, particularly in the context of Kashmir, due to the continuous unrest and violence that have impacted the mental well-being of its inhabitants. Thus, socio-political and cultural factors in Kashmir impact individuals' experiences with mental health and support-seeking.

2.3 Crisis informatics and HCI

Different populations and communities across the globe continue to face disruption to various events such as natural calamities, public health crises, war & violence, etc. The prevalence of various technologies such as ICTs, social media, etc., has also found its usage during such disruptions [43]. Scholarship in HCI, CSCW, ICTD has explored the role and application of technology during crises and disruptions [30, 41, 42], also termed as crisis informatics. Research in crisis informatics has provided heterogeneous account of technology use and implications during various crisis events, such as, floods [41], earthquakes [42], riots, criminal and political violence [10], war [30], etc. Prior literature has shown the different use of technologies during disasters and natural calamities. Shaw et al., [41] in their investigation of Twitter use during Queensland flood

highlighted the use of social media as both communication and an emergency management tool. They found out that social media provided a space for emotional exchanges and distress communication. Similarly, Vieweg et al., [51] studied the use of micro-blogging web posts during the Oklahoma Grassfires of April 2009 and the Red River Floods. They highlighted the need to better design the information extraction systems to enhance situation awareness with a broadcast message for a broader population and individuals. Researchers have also explored the technology use during the public health crisis as well [17]. Gui et al. [17] conducted a qualitative analysis of social media posts related to Zika virus spread to understand the use of such communication by public health bodies and citizens. They unfurled the discussion on technology use in decision-making during such a crisis, as users often struggled to find credible information.

Researchers have also explored the technology used during the conflict, war, and riots [10, 30]. Mark et al., [30] conducted an ethnographic interview study in the context of Iraq that has been facing a protracted conflict to understand the resilient practices adopted by the citizens using technology to carry out their day-to-day activities. Their work highlighted the limitations posed by the environment and how the citizens reappropriated tech, devised patterns, and adopted practices that enabled them to adapt to the environment and even be productive. De Choudhury et al., [8] studied the affective dimensions of Twitter data during the Mexican drug war violence. Their findings highlighted the gradual effective change in the tweets during the protracted war and its impact on people's psychological well-being, and they argue for better designing tools to cater to the mental health needs of such communities. However, the experiences and the challenges cited in the existing scholarship are materially diverse depending on the nature of the conflict and do not generalise uniformly across geographies. Moreover, conflict-affected contexts such as Kashmir still are understudied within HCI research.

3 RESEARCH GOALS AND SUMMARY

My dissertation is anchored in the domain of HCI and particularly focuses on regions marked by socio-political unrest, using Kashmir — a region with conflict for over three decades—as our case study. Through qualitative research practices, my work explores the barriers to various critical domains essential for human development, e.g., education, job-seeking opportunities, entrepreneurship experiences, mental health, and well-being of the people in the region. With this, my work will aim to answer intertwined research questions with an overarching research agenda that aims to understand the “*use of ICTs during ongoing protracted socio-political disruptions*” drawing from case studies across various essential human developmental domains of life across Kashmir. Below, I briefly present the works done and contributions made so far in the domain of education, employment, and mental health and well-being —

Case Study I: Education

RQs: *How do students, parents, and informal educational spaces support and continue education during disruption? What are the different technologies used to impart education, and how are they used during ongoing disruption?*

Sustainable Development Goal 4 promotes inclusive and equitable quality education and lifelong learning opportunities for all. However, regions with ongoing socio-political conflict suffer disruption to education and learning. Situating this work in Kashmir, India, affected by socio-political conflict, I partnered with an NGO in Srinagar, working with vulnerable children and marginalized youth through their quality education and socio-economic initiatives. They have worked actively for over ten years with under-resourced communities in Kashmir, primarily in rural settings. After the abrogation of section 370 in August 2019, they established Community Learning Centers (CLCs) to continue supporting the education of students during the shutdown. These centers aim to prepare students to reintegrate into formal learning environments, including academic sessions, remedial sessions, and recreational activities. I did multiple field visits and conducted 21 semi-structured interviews with parents, teachers, students, and members of the NGO that runs Community Learning Centers in Kashmir. The findings from this work present the barriers in education caused by disruption and the role of community learning centers in overcoming the barriers within these contextual constraints. Further, this work discusses engaging researchers and policymakers to leverage human infrastructure, embedding uncertainty into the design, structuring trust, and content usability to develop solutions to make education more accessible. Despite significant research in HCI and Education, research in this particular context is underexplored, and the work contributes to filling this gap. The work is published at CHI 2022 [54].

Case Study II: Employment

RQs: *How do the existing infrastructure offer support to the youth in the job-seeking process? What are the resources/mentorship available, and how are technological interventions supporting these practices?* The issue of youth unemployment is a global concern, with active initiatives, policies, and technologies in place to support employment seekers. Despite these efforts, youth unemployment in various regions is attributed to the complex challenges that impede access to resources, particularly in contexts characterized by socio-political unrest. To understand the use of job-seeking resources in such a context, situating this work in the region of Jammu & Kashmir, this work employed qualitative methods, I conducted 21 semi-structured interviews (11 male, 10 female) with participants who had recently secured a job or were actively seeking one. The findings from this work present the systemic infrastructural challenges the youths face due to the ongoing crisis. Analyzing the youths' workarounds to mitigate these challenges. This work discusses how we might leverage these workarounds into designing systems while navigating through the contextual complexities. We discuss the challenges of designing in the context and provide design recommendations to support job seekers. The work is accepted at CSCW 2024.

Case Study III: Mental Health and well-being

RQs: *What are the experiences of the people seeking professional mental health services, barriers, and facilitators that exist in accessing resources in Kashmir? How does the use of technology impact the experiences of the people in Kashmir seeking mental health services?*

Mental health well-being is a global concern, with disparities in treatment services being a challenge. Digital mental health interventions are proposed to bridge the gaps and supplement support and assistance. However, many individuals still struggle with mental health issues, particularly in regions encountering socio-political unrest, and face obstacles in seeking professional assistance. Situating this work in Kashmir, India, I conducted 18 semi-structured interviews with participants seeking professional support facilitated by a practicing psychologist in a clinical setting. This work aimed to explore how individuals navigate mental health and professional help-seeking. The findings identify the struggles in seeking support rooted in the context through socio-political and socio-cultural influences, strategies and methods adopted to navigate these struggles, and the role of technology in seeking support. Using a social-ecological approach to mental health care, this study emphasizes accounting for the socio-political realities that shape support-seeking in politically disturbed contexts and offers socio-technical design recommendations. The work is under submission.

4 RESEACH CONTRIBUTIONS AND NEXT STEPS

My research extends the scope of Human-Computer Interaction (HCI) scholarship by providing empirical insights into the technological infrastructure in varied critical developmental areas of life (e.g., education, employment, mental health, and well-being) in Kashmir, India. This dissertation contributes to a deeper understanding of the context and lays the groundwork for future interventions with conflict-affected communities. This work contributes to the HCI and HCI for development towards the designing of socio-technical systems to address technological failures within such contexts. My work expands the conversation and understanding of the HCI community. It emphasizes that it is of utmost importance for researchers to engage with conflict-affected communities across the globe to understand critical design challenges, which are often shadowed by the mainstream progression and assumptions around technologies.

Furthermore, based on the observations and findings from my prior studies, I identified that gender and technology use have a different interplay in our context for men and women and are shaped by socio-political and socio-cultural values. With these observations, I propose a study as Phase 2 of my research that focuses on understanding the patterns or challenges with entrepreneurship journeys among women within both online and offline spaces, more specifically, as two studies, respectively. For online entrepreneurship, I aim to answer the research question that focuses on understanding women's entrepreneurship experiences and the challenges and stigmas they face. And how is the technological infrastructure supporting that? Similarly, for offline entrepreneurship experiences, I would focus on answering the women's entrepreneurship experiences, motivation, and the challenges and stigmas they face. And how is the technological infrastructure supporting? Further, based on the learnings from these studies, I plan to engage in speculative and co-design sessions with the entrepreneurs and understand what the system's design might look like. For these studies, I plan to use qualitative and design methods. More specifically, the studies will draw on semi-structured interviews with women entrepreneurs in

Kashmir (both online and offline). The studies would recruit the participants using online mediums (e.g., social media platforms). I would also engage with an NGO that works on women's empowerment and skill development initiatives to promote their economic participation and address the socioeconomic and cultural barriers they face. Further, the data analysis will involve a thematic analysis of the interviews and a systematic literature review.

5 BIO AND DESSERTATION STATUS

I am a PhD student in Computer Science and Engineering Department at the Melange Lab, IIIT Delhi, advised by Professor Pushpendra Singh. I am currently in my fifth year, and I anticipate completing my PhD by January 2025 at the latest. I plan to initiate Phase 2 of my proposed research in the first half of 2024. Participation in the doctoral consortium will provide valuable insights into the co-design aspects of my work, as well as guidance on approaching the dissertation writing based on the broad array of case-based studies in my research work. Upon completing my PhD, I aspire to secure research positions in either industry or related academic departments.

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